Library Management Book Availability System

Project Report

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Project overview

A course prototype combining historical borrowing analysis with a database-backed inventory dashboard. It helps readers find titles and staff review circulation and collection status.

Problem statement

Readers and staff need a clear way to check whether books are available, reserved, borrowed or overdue. When inventory and borrowing information are difficult to scan together, readers may not know which title to choose and staff spend time answering routine status questions.

Proposed solution

Preserve and clean the source CSV, analyze borrowing patterns and create charts. Provide an interactive dashboard that reads book inventory and active-loan records from SQLite to show copy-level availability and support borrowing and returns.

How it helps users

- Students can search titles and see available-copy counts before choosing a book.
- Library staff can record borrowing and returns, review loan history and identify overdue loans.
- Administrators can compare collection size, category mix, demand and usage when planning purchases.

Project files and GitHub workflow

File or folder	Purpose
data/raw/library_datasets.csv	Original Kaggle dataset; kept as the source copy.
data/processed/library_datasets_cleaned.csv	Cleaned 1,000-row, 9-column dataset for record-based analysis.
data_cleaning.py	Normalizes fields, removes exact duplicates, parses dates and standardizes known flags.
Data_analysis.py	Writes category, reservation, overdue, duration and monthly borrowing summaries.
Data_visualisation.py	Creates category, reservation, overdue and borrowing-duration charts from the cleaned CSV.
reports/	Analysis summary plus saved chart images.
dashboard/app.py	Main Streamlit dashboard with five navigation sections.
dashboard/catalogue.py / loans.py / live_inventory.py	Focused Streamlit modules for catalogue, loan actions and inventory views.
database/ and library.db	SQLite schema/seed helpers and the project inventory/loan database.
webapp/backened/main.py	API backend source.
Dashboard.pbix	Power BI project file is present in the repository.
README.md / PROJECT_PROPOSAL.md / TASK_ASSIGNMENTS.md	Run instructions, project scope and task-to-issue mapping.

Data workflow

Kaggle CSV > raw file > cleaning > analysis > charts > dashboard. The CSV describes historical events; SQLite books and loans provide current copy-level availability in the application.

GitHub collaboration

- Issues #10–#15 track proposal, dataset collection, cleaning, dashboard, analysis and visualisation tasks.
- Merged pull requests on main: #16 dashboard, #17 dataset collection, #18 borrowing analysis. The checked-out main branch contains these merges.
- MISSING_ITEMS_CHECKLIST.txt still says Dashboard.pbix is missing although the file is present. Update the checklist before submission.

Repository: github.com/saranyaachowdary027/Library-management-book-availability-system

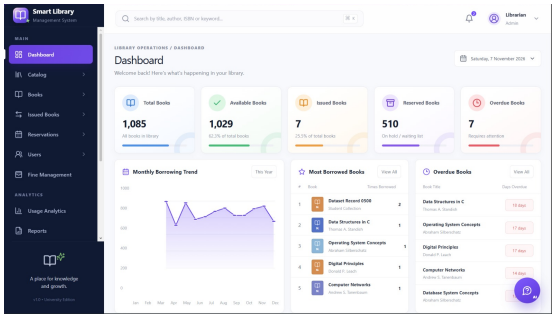
Power BI dashboard Dashboard.pbix

The repository includes a separate Power BI report. Its single page is titled “Smart Library Management System” with the subtitle “Borrowing & Reader Insights.”

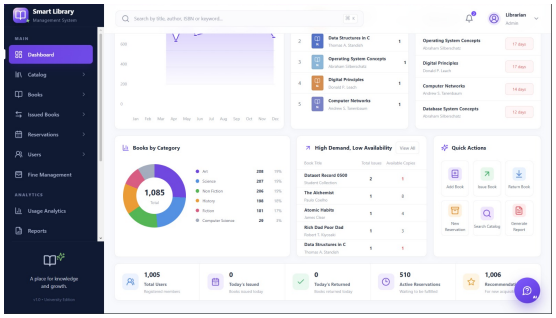
- KPI cards are labeled Total Borrowing Records, Unique Readers, Overdue Loans and Reserved Transactions.
- Slicers filter by reservation status, borrow-date range and user role.
- Visuals include borrowing records by book category, yearly borrowing trend, reservation and overdue breakdowns, borrowing by user role and borrowing duration by role; a category detail table is also included.
- Use the PBIX to filter and compare historical borrowing patterns. It is separate from the Streamlit screenshots and is not live copy-level inventory.

Full dashboard walkthrough

The main app is dashboard/app.py. Its sidebar has five sections. Inventory and loan screens read SQLite; Historical Borrowing Analytics reads the cleaned CSV.



Overview cards, availability trend and circulation panels



Category, demand, title lists and quick actions

1. Overview

- Cards: book titles, total copies, available copies and currently borrowed.
- Bar chart compares available and total copies by title; circulation status shows overdue loans, titles with no copies available and active loans.
- Current-inventory table plus quick links to Catalogue, Borrow & Return and Live Inventory.

2. Live Inventory

- Search by title, author or category. Cards show matching titles, available copies and unavailable copies.
- Table lists book ID, title, author, category, total copies and available copies.

3. Catalogue

- Search the collection by title, author or category; review matching titles with metadata and copy counts.

4. Borrow & Return

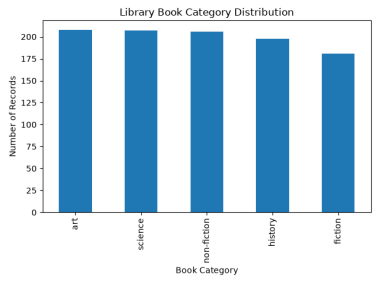
- Borrow tab: select an available title, enter borrower name, set borrow and due dates; the app checks availability and date order.
- Return tab: select an active loan, enter the return date and confirm. Loan history tab lists loan records and returned/borrowed status.

5. Analytics

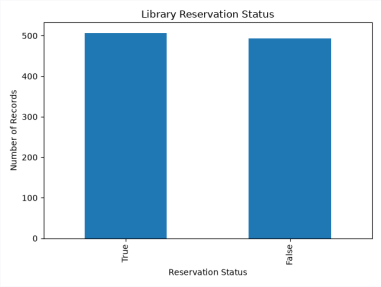
- Shows the number of historical CSV records, a category bar chart, a monthly borrowing line chart and the full records table.
- This page is separate from current SQLite inventory; it summarizes historical borrowing activity.

Charts, findings and interpretation

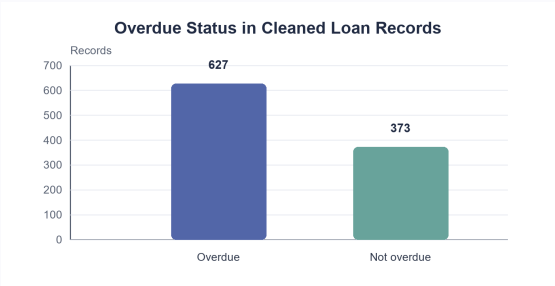
The chart script creates record-based visualisations from the cleaned CSV. The overdue chart below is rendered from the same cleaned records for this report.



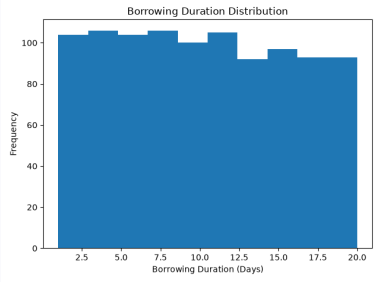
Category records: Art 208, Science 207, Non-fiction 206, History 198, Fiction 181.



Reservations: 507 true and 493 false.



Overdue flag: 627 overdue, 373 not overdue.



Loan duration: mean 10.22 days, median 10, range 1–20.

Key interpretation notes

- Category groups are close in size (181–208 records); reservation flags are nearly balanced.
- popular_books.png lists book IDs seen once each, so this CSV cannot identify repeat-borrowed or genuinely popular titles.
- book_availability.png bars show reservation True/False; clarify its filename before presenting it as availability data.
- The cleaned CSV has 1,000 historical loan rows, not a live stock feed. Current availability uses inventory and unreturned SQLite loans.
- Dashboard screenshot counts show 1,029 available of 1,085 total (94.8%), but the image label reads 62.3%. It shows 7 issued (0.6%) while the label reads 25.5%. Verify percentages before presenting them.

Conclusion

The project combines a repeatable data workflow, interpretable charts and an interactive dashboard. Historical CSV analysis and current database availability answer different questions; keep those sources distinct when describing results.